

Model	Input	Purpose	Output	Evaluation
M0 Raw TF-IDF	Original module → text and staff profiles	Create a transparent lexical baseline	Ranked staff candidates	Case study; NDCG at 5, precision, MRR and MAP
M1 Enriched TF-IDF	M0 data plus → publication metadata	Test whether enrichment improves lexical matching	Updated ranked candidates	Change from M0; benchmark ranking metrics
M2 Semantic model	Cleaned module → text and enriched staff profiles	Capture contextual similarity beyond keywords	Semantic ranking; M2-H eligibility gate	Coverage, rejection audit and benchmark metrics
M3-KG Graph hybrid	M2-H candidates → plus topic and organisational graph	Rerank candidates and explain recommendations	Reranked list and evidence paths	Rank changes, path coverage, topic evidence and NDCG
M4 Collaboration	Canonical staff → and module topic profiles	Find complementary cross-department staff pairs	Ranked primary and guest-collaborator pairs	Combined topic coverage, complementarity gain and retained pairs
M5 Capacity-aware	Candidate lists → plus simulated workload constraints	Reduce assignment concentration while retaining relevance	Simulated balanced allocation	Gini, utilisation, maximum load and relevance retention